

4	(a) (i) solid has fixed volume and fixed shape/incompressible	B1	[1]
	(ii) gas fills any space into which it is put	B1	[1]
	(b) atoms/molecules have (elastic) collisions with the walls (of the vessel) momentum of atom/molecule changes <u>so</u> impulse (on wall)/force on wall random motion/many collisions (per unit time) gives rise to (constant) force/pressure	B1 B1 B1 B1	[4]
	(c) spacing (much) greater in gases than in liquids/about ten times <i>either</i> spacing depends on $1/\sqrt[3]{\rho}$ <i>or</i> ratio of spacings is about 8.8	C1 A1	[2]
5	(a) (i) 1 number of oscillations per unit time (not per second)	B1	[1]